What Does Conformal Prediction Add?

A Controlled Audit of Calibrated Set-Valued Classification

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Abstract. A conformal predictor comes with a finite-sample coverage guarantee. The number a pipeline reports does not. We audit three choices that sit between the guarantee and the reported number: the calibration of the threshold, the treatment of empty sets, and the update rule of adaptive conformal inference. Across two image datasets, a tabular benchmark, and a range of model strengths, a fixed probability threshold is not a coverage method, a threshold calibrated on the true-class probability is exactly split conformal, and forcing the argmax into empty sets raises the reported coverage by up to 9.7 points at $\alpha = 0.10$ on the image benchmarks. On a tabular benchmark the same mechanism is near zero, as the empty-rate diagnostic predicts. The common α_t update is not ACI and reports higher coverage by inflating sets. The audit of eight libraries shows that the empty-set policy varies, that one widely used default over-covers by 4.6 points, and that three ACI implementations use the correct fixed-target update. We organise the field by guarantee and assumption rather than by chronology, and we give a six-item checklist. The guarantee constrains the procedure, not the number that the pipeline prints.

Keywords. conformal prediction; coverage; calibration; post-processing; adaptive conformal inference; implementation audit.

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1. Introduction

1.1 A number and a guarantee

A practitioner sets a 0.90 coverage target. The simplest way to meet it is to threshold the softmax at 0.90, keep the labels above the threshold, and report the fraction of test points on which the true label is in the set. The reported number is usually close to the model's accuracy, and it is often presented as evidence that the model is calibrated. It is not. The threshold is a fixed probability, not a calibrated quantile, and the set it produces is a singleton or empty. The number is a property of the threshold rule, not of the guarantee.

The conformal alternative replaces the fixed threshold with a quantile of the calibration scores. The resulting set satisfies a finite-sample coverage guarantee under exchangeability [38, 3]. The guarantee is real, and it is the reason the method is worth using. The guarantee is also a statement about the procedure, not about the number a particular pipeline prints. Between the procedure and the number sit at least three choices: how the threshold is calibrated, what the implementation does with an empty set, and how an adaptive method updates its target. Each choice can move the reported coverage without changing the guarantee. This paper is about those three choices.

1.2 Three places where the number moves

The first is the threshold. A fixed probability threshold is not a coverage method. It returns a singleton for the labels above the threshold and an empty set for the rest, and its coverage is the model's accuracy on the points it does not abstain on. The correct calibration is a threshold on the true-class probability, and that threshold is exactly split conformal with the naive score. The identity is one line, and it is known [1]. We do not claim it. We measure what follows from using the wrong threshold. On CIFAR-10, a raw 0.90 threshold returns an empty set for 24.4 percent of the test points even when the model is 85.5 percent accurate, and a threshold calibrated on the maximum probability still falls short of the target.

The second is the empty set. A conformal set can be empty, and many implementations insert the argmax so that the output is never empty. The insertion is a post-processing rule, and it changes the reported coverage by the empty rate times the conditional accuracy of the argmax on the empty sets. At $\alpha = 0.10$, the change reaches 9.7 points on digits and 9.5 on CIFAR-10. The effect is not a property of the score. It is a property of the interaction between the score and the post-processing, and it can turn a method that is on target into one that over-covers by several points.

The third is the update rule. Adaptive conformal inference moves the target level after each observation. The published recursion uses a fixed target, and a common implementation replaces it with the current level. The replacement removes the restoring force toward the nominal level, and the wrong rule can report a higher coverage by driving the level to the lower clip and inflating the sets. On CIFAR-10 the wrong rule reports 10.8 more points of coverage than the correct one, at a 41 percent larger mean set size. The correct rule is not a cure-all. On the streams we tested it also falls short of the target, for reasons that the current theory does not separate.

The three choices are not independent. The first decides whether a comparison is between two rules or between two calibrations. The second decides whether a comparison is between two scores or between two post-processing rules. The third decides whether an adaptive method is adapting or inflating. A paper that ignores all three can report a number that is internally consistent and externally meaningless.

1.3 What this paper does

We audit the three choices on two image datasets and a tabular benchmark. The digits benchmark is small enough to check by hand. CIFAR-10 is large enough that the coverage estimates are tight. Covertype is a tabular benchmark with seven classes, used to test whether the post-processing effect survives a different modality. We vary the score, the target level, the calibration size, and the model strength, and we evaluate every cell twice, once with the empty sets kept and once with the argmax forced. The image grid has 1,260 cells; the tabular replication adds 180 rows. We also audit eight libraries, all at the source level and five at runtime, and we compare their empty-set policies and their update rules.

The audit is organised by a two-axis map. The rows are the guarantee a method pursues: marginal coverage, group-conditional coverage, pointwise conditional coverage, and risk control. The columns are the assumption it is willing to make: exchangeability, covariate shift, non-exchangeability, and known groups. The map replaces the chronology that a survey usually imposes, and its empty cells are the open problems of Section 7.

1.4 Contributions and positioning

The paper makes four contributions. First, the map, which makes the assumptions explicit and the costs comparable. Second, the audit, which measures the three choices across two image datasets and a tabular benchmark, and separates the raw from the forced evaluation. Third, the library policy map, which shows that the empty-set policy varies, that one widely used default over-covers by 4.6 points, and that three ACI implementations use the correct fixed-target update. Fourth, a six-item checklist for practitioners.

Four surveys are close enough to require positioning. Angelopoulos and Bates [3] is a tutorial. It builds the reader's intuition for the split construction and covers the pre-2022 literature. Fontana et al. [18] is a unified review of the theory. Campos et al. [12] organises the NLP literature by task, and Zhou et al. [39] organises the field by data modality. None of them audits the implementation choices that this paper measures. The closest 2025-2026 work is narrower. Aguru [1] proves the binary threshold geometry that our first audit uses. Mehrtens et al. [27] document a different set of pitfalls in medical imaging. Gauthier et al. [19] treat empty sets as a low-coverage outcome. Al Bataineh [2] uses empty sets as a drift alarm. Boman [11] benchmarks score functions without measuring the post-processing. The audit complements these papers rather than repeating them. Preprints are marked in the reference list, and the software versions are listed there as well.

1.5 Scope

The paper is an audit, not a benchmark. The datasets are three, the models are linear heads, and the drift is a synthetic ramp on digits and a corruption family on CIFAR-10. The claims are about mechanisms, not about the magnitudes that a deployment should expect. The bibliography is a curated pool, not a systematic census; the cited subset is what the argument needs. The empty cells of the map mean "no method in the reviewed set," not "no method can exist." A reader who wants a positive result about adaptive inference will not find it here. A reader who wants to know how the reported number can move without the guarantee moving will.

The rest of the paper is organised as follows. Section 2 states the guarantee and the finite-sample correction. Section 3 introduces the map. Section 4 audits the three choices and the libraries. Section 5 describes the experiments. Section 6 gives the checklist. Section 7 states the open problems.

2. The guarantee in one page

2.1 Exchangeability and what the guarantee assumes

A conformal predictor is a procedure, not a model. It takes a fitted classifier f, a calibration sample, and a target error rate α, and it returns a set C(x) for each input. The only assumption is that the calibration pairs and the test pair are exchangeable. Exchangeability means that the joint distribution is invariant under permutation of the indices. It is weaker than independence and identical distribution. Every i.i.d. sample is exchangeable, and so is any sample whose distribution is symmetric in the order of the observations.

The assumption is easy to state and easy to violate. A calibration set drawn at random from the same collection as the test set is exchangeable with it. A calibration set drawn from last year's data and deployed against this year's data is not. A calibration set selected because it contains the cases the model finds hard is not. A test stream whose distribution changes after every observation is not. In each case the guarantee does not degrade gracefully. The coverage on the deployment distribution can fall far below $1 - \alpha$ while every diagnostic computed on the calibration distribution looks correct. This is the failure mode that Section 5.4 measures.

The guarantee is marginal in a second sense as well. It averages over the randomness of the calibration sample and the test point. It does not say that a particular realized set covers the particular test label. It says that the procedure covers with probability at least $1 - \alpha$ over repeated draws. A single calibration set can produce coverage below $1 - \alpha$, and the probability that it does is not zero. For a binary classifier, the calibration-conditional coverage follows a Beta law, and the probability of a shortfall can be computed exactly [1]. The practical consequence is that a single estimate of 0.8996 against a target of 0.90 is not evidence of a bug, and a single estimate of 0.946 is not evidence of a better method. Both are draws from a procedure whose average is fixed.

2.2 Split conformal

Figure 1 shows the three stages (solid) and the reporting step (dashed), with the three places where the reported number can move. The construction is the split conformal predictor of Vovk et al. [38] and Shafer and Vovk [33]. Fit f on a training set. Pass a disjoint calibration set of size n through f and compute a nonconformity score $s_i = s(X_i, Y_i)$ for each calibration pair, where a larger score means the pair is less consistent with the model. Sort the scores and take the k-th smallest, with $k = \text{ceil}((n + 1)(1 - \alpha))$. For a new input x, the prediction set is the collection of labels y whose score does not exceed q.

The index k is the part that is most often implemented incorrectly. The naive choice $\text{ceil}(n(1 - \alpha))$ gives a threshold that is too small, and the resulting coverage is below the nominal level by a term of order 1/n. The choice $\text{ceil}((n + 1)(1 - \alpha))$ costs one effective calibration sample and buys a guarantee that holds for every n, not only in the limit. With continuous scores, the coverage is exactly $k/(n + 1)$, which exceeds $1 - \alpha$ by at most $1/(n + 1)$ [38, 24]. The correction is not cosmetic. It is the difference between a procedure with a finite-sample guarantee and a procedure with an asymptotic one.

The guarantee is a lower bound, and the upper bound is also informative. At n = 100 and α = 0.10 the coverage is 91/101 = 0.901. At n = 10 it is 10/11 = 0.909. The absolute difference is small, but it is the difference between a guarantee that is tight at the nominal level and one that is visibly conservative. The calibration size is therefore a design variable, not a nuisance. Section 5 varies it explicitly.

2.3 Scores and efficiency

Once the quantile is fixed, the only design freedom is the score. For classification, the simplest choice is $s(x, y) = 1 - p_y(x)$, which yields the set of labels whose predicted probability exceeds a threshold. This is the least-ambiguous classifier of Sadinle et al. [32], and it is the score that split conformal reduces to when the threshold is chosen on the calibration set. The score is global. A class that the model systematically under-predicts is excluded more often than the nominal rate allows, and the marginal coverage is restored only by compensation across classes.

Adaptive scores change the object being thresholded. The APS score accumulates probability mass in descending order until the true label is reached, so the score depends on the rank of the true label rather than on its absolute probability [31]. The randomization in the score is what restores exact validity in the presence of ties. The regularized variant adds a penalty on the number of labels in the set and shrinks large sets at a small cost in coverage [6]. For regression, the analogous choice is the absolute residual, and the conformalized quantile regression of Romano et al. [30] uses a locally adaptive residual.

Efficiency has no distribution-free optimum. There is no score that is uniformly smallest across distributions, and the achievable set size depends on the model and on the data. A better model produces smaller sets, and a worse model produces larger ones, but neither changes the validity of the procedure. This is why the efficiency literature is a sequence of score designs rather than a sequence of theorems, and why the choice of score is an empirical question. It is also why the score determines the empty-set behavior that Section 4.2 measures. A score that can place every label above the threshold produces empty sets. A score that cannot produces sets that are never empty. The validity guarantee is silent on which case obtains.

2.4 Marginal and conditional coverage

Marginal coverage averages over the covariate distribution. Conditional coverage would require $P(Y \text{ in } C(X) \mid X = x) \geq 1 - \alpha$ for almost every x. The second statement is much stronger, and it is not attainable without assumptions. Barber et al. [7] show that no distribution-free procedure can guarantee pointwise conditional coverage with non-trivial sets for all distributions. The intuition is that a finite calibration sample contains finite information about any particular x, and a procedure that must be valid for every distribution can always be confronted with a distribution that concentrates its mass elsewhere. In classification, the only procedure that attains the target everywhere is the trivial one that returns the entire label space.

The field's responses weaken the target rather than the impossibility. Group-conditional coverage requires the guarantee to hold on average within each of a collection of groups. When the groups are known in advance, Mondrian calibration attains it by calibrating separately within each group [37]. The cost is a smaller effective calibration sample per group and the requirement that the group label be available at test time. Local weighting attains a smoother version of the target by assuming that the conditional coverage function varies slowly in x, and randomization makes the local weights identifiable [22]. The assumption is explicit, and it is not verifiable from data alone.

Class-conditional coverage is a group-conditional target, not a pointwise one. A class-conditional coverage below the nominal level for the worst-served class is a statement about a known group, and it is attainable with Mondrian calibration. It is not evidence for or against the pointwise impossibility of Barber et al. Treating the two as the same is a category error. The distinction matters for Section 5, where the class-conditional deviations are large while the marginal coverage is nominal.

2.5 What the guarantee does not say

The guarantee does not say that the set is small. It does not say that coverage holds for every subgroup. It does not say that the calibration set is representative of the deployment distribution. It does not survive non-exchangeability without modification. It is a statement about a procedure, not about a particular realized set, and it is a lower bound on an average, not a promise about the next prediction.

These limits are not defects of the theory. They are the terms of the contract. The rest of the paper is about what happens when the terms are forgotten. A reported coverage number is the output of a pipeline that includes a score, a calibration set, a threshold rule, and a post-processing rule. The guarantee constrains one of those objects. It does not constrain the number that the pipeline prints.

3. A two-axis map

3.1 Why a map, not a chronology

A chronological account of conformal prediction is easy to write and hard to use. The field did not develop in phases. The split construction, the adaptive score, the impossibility result, and the risk-control extension were motivated by different problems, and several of them overlap in time. A reader who memorizes the chronology learns a list of methods. A reader who follows the guarantee learns where each method breaks.

The map in Table 1 has two axes. The rows are the guarantee a method pursues: marginal coverage, group-conditional coverage, pointwise conditional coverage, and risk control. The columns are the assumption it is willing to make: exchangeability, covariate shift, non-exchangeability, and known groups. A method is located by the pair. A method that claims a guarantee outside its column is mis-specified, and a method that is valid in one column need not be valid in another.

The axes are not independent. Moving right weakens the data assumption. Moving down strengthens the guarantee. The two directions compete: a stronger guarantee usually requires a stronger assumption, a larger set, or both. The empty cells are where the competition has not been resolved.

3.2 The rows

The first row is marginal coverage. The guarantee is $P(Y \text{ in } C(X)) \geq 1 - \alpha$, averaged over the data distribution. It is the base case, and it is the only case that is attainable without assumptions beyond exchangeability. The split construction of Vovk et al. [38], the regression analysis of Lei et al. [24], and the tutorial of Angelopoulos and Bates [3] all live here. The cost is one calibration pass, and the score is free. The guarantee is weak in the sense that it says nothing about any particular x. It is strong in the sense that it survives a badly misspecified model.

The second row is group-conditional coverage. The guarantee is that coverage holds on average within each of a collection of groups. When the groups are known in advance, Mondrian calibration attains it by calibrating separately within each group [37]. Gibbs et al. [21] develop a framework for group-conditional coverage under weaker assumptions, and the cost is the same in kind: a smaller effective calibration sample per group, and the requirement that the group label be available at test time. The guarantee is stronger than marginal coverage and weaker than pointwise coverage. It is the right target when the deployment has a known partition, such as a demographic group or a class label.

The third row is pointwise conditional coverage. The guarantee is $P(Y \text{ in } C(X) \mid X = x) \geq 1 - \alpha$ for almost every x. Barber et al. [7] show that no distribution-free procedure can attain it with non-trivial sets for all distributions. The constructive route adds assumptions. Hore and Barber [22] assume that the conditional coverage function varies slowly in x, and they show that randomization makes the local weights identifiable. The assumption is explicit, and it is not verifiable from data alone. The sets are larger than the marginal ones, and the guarantee is conditional only up to the smoothness constant.

The fourth row is risk control. The guarantee is $E[L(C(X), Y)] \leq r$ for a bounded loss L, of which coverage is a special case. Bates et al. [9] introduce the general construction, and later work extends it to a grid of risk levels with multiplicity control [4] and to a cross-validated version that removes the data-splitting loss [13]. The assumption is a monotone loss and exchangeability. The cost is a grid search over the risk level and a correction for the number of candidates. The guarantee is not a coverage statement, and it does not reduce to one unless the loss is the indicator of a miss.

3.3 The columns

The first column is exchangeability. It is the base assumption, and every method in the first column is valid under it. The column is not a statement about the data. It is a statement about the procedure's tolerance. A calibration set drawn at random from the same distribution as the test set satisfies it. A calibration set drawn from a different distribution does not.

The second column is covariate shift. The guarantee is recovered by reweighting the calibration scores according to the likelihood ratio between the test and calibration covariate distributions [35]. The assumption is that the likelihood ratio is known or estimable. When it is estimated, the estimation error propagates into the guarantee, and the coverage is exact only if the ratio is exact. The column is therefore narrower than it looks. It is not a distribution-free guarantee under arbitrary shift. It is a guarantee under a shift that can be quantified.

The third column is non-exchangeability. The guarantee is weakened to a bound that depends on a measure of the violation. Barber et al. [8] give a bound in terms of a total-variation-like quantity, and Oliveira et al. [28] study the split construction under a known ordering. The bound is exact, but the quantity is not estimable from data in most applications. The alternative is online adaptation. Gibbs and Candès [20] update the target level after each observation, and the guarantee becomes an average over time rather than a statement about each step. The distinction matters. A per-step guarantee is not available without assumptions. An average guarantee is, and it is what the adaptive methods provide.

The fourth column is known groups. The method is Mondrian calibration, and the guarantee is group-conditional [37]. The assumption is that the group label is known at calibration and at test time. The cost is a smaller effective sample per group, and the guarantee is only as good as the group definition. A group that is defined by a variable the model can see is easy. A group that is defined by a variable the model cannot see is not.

3.4 The empty cells

The map has empty cells, and they are the point. Pointwise conditional coverage under covariate shift is empty in the reviewed set. Pointwise conditional coverage under non-exchangeability is empty. Group-conditional coverage under covariate shift and under non-exchangeability is empty. Risk control under covariate shift is empty in the reviewed set. These are not claims that no method can exist. They are claims that no method in the reviewed set occupies the cell, and they are the open problems of Section 7.

The empty cells also explain why the field's arguments often talk past each other. A method that is valid in the exchangeable column can be presented as if it were valid under shift, and a reader who does not check the column will believe it. The map makes the check explicit. It is the first question a practitioner should ask of a new method. Which cell does it occupy, and what does it assume to get there?

3.5 What the map buys

The map buys three things. First, it makes the assumptions explicit. A method's guarantee is a conditional statement, and the condition is the column. Second, it makes the costs comparable. A group-conditional guarantee costs a smaller effective sample. A pointwise guarantee costs an unverifiable smoothness assumption. A risk-control guarantee costs a grid and a multiplicity correction. Third, it makes the empirical audit interpretable. A reported coverage number is a property of a cell, not of a score. The same score can produce 0.90 in one cell and 0.95 in another, and the difference is the post-processing, not the guarantee.

Table 9 places the map against four existing surveys. The comparison is about coverage, not quality. The map's contribution is the two-axis organization and the explicit empty cells. The empirical contribution is the audit of whether the reported numbers respect the cell. Neither is a substitute for the other.

4. Three implementation traps

4.1 A calibrated threshold is split conformal

The simplest baseline for a coverage target is a probability threshold. Choose $t = 1 - \alpha$, keep the labels whose predicted probability exceeds t, and report the empirical coverage on a test set. The procedure is one line of code, it is available in every classifier, and it is not a coverage method. The audit in this section shows why, and it shows that the correct version of the baseline is exactly split conformal.

The identity is one line. For the naive score $s(x, y) = 1 - p_y(x)$, the conformal set is $\{y : 1 - p_y(x) \leq q\} = \{y : p_y(x) \geq 1 - q\}$, where q is the k-th smallest calibration score. The right-hand side is a threshold rule with $t = 1 - q$. The conformal quantile is the $(1 - \alpha)$ quantile of $1 - p_true$, so $1 - q$ is the corresponding quantile of the true-class probability, and the calibrated threshold is the same object. The set construction is not doing any work beyond the calibration of the threshold. This is a known consequence of the score definition, and for binary classifiers the threshold geometry has been stated explicitly [1]. In multi-class classification it is an immediate corollary of the probabilities summing to one. We do not claim the identity. We measure what follows from ignoring it.

The measurement uses four rules at $\alpha = 0.10$. The raw threshold fixes $t = 0.90$. The max-probability rule chooses t as the α-quantile of the maximum predicted probability on the calibration set. The true-class rule chooses t as the α-quantile of the true-class probability. The conformal rule uses $t = 1 - q$. The first is the baseline a practitioner writes. The second is the repair a careful practitioner tries. The third is the correct calibration. The fourth is split conformal.

Table 2 reports the digits results at $n_cal = 449$, and Table 5 reports CIFAR-10 at $n_cal = 2000$, across four training-set sizes. Three facts are stable across the two image datasets.

First, the raw threshold is degenerate. At $\alpha = 0.10$ and a training set of 44 digits, the raw threshold returns an empty set for 70.1 percent of the test points, and its coverage is 0.294. Forcing the argmax into the empty sets raises the coverage to 0.833, which is the model's accuracy. On CIFAR-10 with 30,000 training images and 85.5 percent accuracy, the raw threshold still returns an empty set for 24.4 percent of the test points, and forcing the argmax raises the coverage from 0.716 to 0.855. The pattern is the same at every strength. The raw threshold is not a weak coverage method. It is a singleton rule with an abstention region, and its coverage is the accuracy of the model on the points it does not abstain on.

Second, the max-probability calibration is closer but not exact. On CIFAR-10 the coverage is 0.668, 0.762, 0.771, and 0.810 at the four strengths. On digits it is 0.800, 0.886, and 0.901. It is close to the target only when the model is accurate enough that the argmax is almost always the true class. The rule calibrates the wrong quantity. The threshold should be chosen on the true-class probability, because that is the event the coverage statement is about. Calibrating the maximum probability answers a different question, and the answer is right only when the two probabilities coincide.

Third, the true-class calibration and split conformal are the same rule. The pointwise set comparison has zero mismatches on 450 test points times five seeds times three strengths for digits, and on 10,000 test points times five seeds times four strengths for CIFAR-10. The two rows in Table 2 and Table 5 are not approximately equal. They are equal. The equivalence is the empirical form of the identity, and it is the reason the correct baseline for a threshold comparison is split conformal.

The practical consequence is a change in what a baseline is. A paper that compares conformal prediction to a raw 0.90 threshold is comparing a calibrated rule to an uncalibrated one, and the comparison is guaranteed to favor conformal for a reason that has nothing to do with set construction. A paper that compares conformal to a threshold

calibrated on the true-class probability is comparing two implementations of the same rule, and the comparison should find no difference. The interesting comparison is the third one: conformal against a different score, or against a different calibration set, or against a different post-processing rule. The first two comparisons are already decided by the algebra.

Figure 2 plots the coverage and the mean set size for the six rules at the three digits strengths. The raw threshold sits at the accuracy floor, the max-probability rule sits between the floor and the target, and the true-class, conformal, and threshold-at-1−q rules sit on top of each other. The figure is a visual proof of the identity, and it is the first thing a reader should look at before reading the rest of the section.

The identity is not a criticism of conformal prediction. It is a clarification of what the method adds. The method adds the calibration of the threshold, not the shape of the set. Once the threshold is calibrated, the set is determined by the score. Section 4.2 is about what happens when the score produces an empty set and the implementation fills it.

4.2 Non-empty post-processing inflates coverage

A conformal set can be empty. The construction does not forbid it. If the score of every label exceeds the threshold, the set has no elements. Some implementations leave the empty set as it is, and some insert the argmax so that the output is never empty. The choice is usually presented as a convenience. It is a post-processing rule, and it changes the coverage of the procedure. This section measures the change.

The accounting is exact. Let E be the set of test points whose raw set is empty, and let a_E be the fraction of those points on which the argmax is correct. Forcing the argmax adds one covered point for each point in E on which the argmax is correct, and no coverage for the others. The change in coverage is therefore |E|/N times a_E, which is the empty rate times the conditional accuracy of the argmax on the empty sets. Write it as $\Delta = e\, a_E$, with $e = |E|/N$. The identity is not an approximation. It is the definition of the two quantities.

The experiment varies the score, the target level, the calibration size, and the model strength. On digits the grid has 540 cells, 4 scores times 3 levels times 3 calibration sizes times 3 strengths times 5 seeds. On CIFAR-10 it has 720 cells, with 4 strengths instead of 3. Every cell is evaluated twice, once with the empty sets kept and once with the argmax forced. The full results are in Table 3 and Table 6, and Figure 3 plots the change against the empty rate.

Across the 540 digits cells the mean change is 0.077 and the largest is 0.318. The mean is 0.041 for the naive score, 0.080 for randomized APS, 0.108 for deterministic APS, and 0.078 for RAPS. The ranking is the ranking of the empty rates. The naive score produces few empty sets because its threshold is low. The deterministic APS produces many because its cumulative score can exceed the threshold for every label. On CIFAR-10 the naive score produces no empty sets at any strength, and the change is zero. The randomized APS produces 1.0 to 3.7 percent empty sets and changes coverage by 0.9 to 3.6 points. The deterministic APS produces 6.9 to 9.5 percent empty sets and changes coverage by 6.8 to 9.5 points.

The change is not confined to weak models. On CIFAR-10 with 100 training images, deterministic APS goes from 0.904 to 0.972, a change of 6.8 points. With 1,000 images it goes from 0.902 to 0.993, a change of 9.1 points. At 30,000 images it goes from 0.899 to 0.994, a change of 9.5 points. On digits the largest cell is 9.7 points at 224 training points. The change does not shrink with model strength, because the empty rate does not shrink. The naive score is the exception: it produces no empty sets on CIFAR-10 at any strength, and the change is zero. The naive score on digits at full strength produces 7.8 percent empty sets and changes coverage by 5.4 points. The empty rate is a property of the score and the calibration, not of the model alone.

The 5,000-image CIFAR-10 cell is the exception that proves the rule. Deterministic APS reaches 0.994 coverage with a mean set size of 6.08 and no empty sets. The score is already conservative, so the fallback has nothing to do. The cell is a reminder that there are two ways to over-cover: a non-empty rule, and a score that is too conservative. The audit isolates the first, but a reader who only looks at the coverage cannot tell them apart.

Figure 3 plots the change against the empty rate for all 1,260 cells. The points lie just below the identity line, and the distance from the line is the conditional accuracy of the argmax on the empty sets. The naive points are furthest below, because the argmax is less often correct when the naive set is empty. The APS points are close to the line, because the argmax is often correct when the APS set is empty. The figure is the visual form of the accounting identity, and it is the reason the change is predictable from the empty rate alone.

The result has a direct consequence for the library audit in Section 4.4. MAPIE's default APS reaches 0.9457 with zero empty sets. The randomized setting reaches 0.8998. The 4.6-point gap is the non-empty rule, not the score. The corrected conclusion is that the default is not the exact conformal rule, and the coverage it reports is the coverage of the post-processing.

The recommendation is not to ban the fallback. An empty set is a valid output, and a deployment may prefer a guess to an abstention. The recommendation is to report both numbers and to compare methods at the same post-processing rule. A comparison between a forced method and an unforced method is a comparison between two procedures, not between two scores. Section 4.4 shows that the choice is exposed in some libraries and hidden in others, and that the default in one widely used library is the non-empty rule.

4.3 The alpha_t update is not ACI

Adaptive conformal inference replaces the fixed target with an online update. After each observation, the procedure observes whether the set covered the true label, and it moves the target level in the direction that would have corrected the error. The update in Gibbs and Candès [20] is $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \mathrm{err}_t)$, where err_t is one if the set missed and zero otherwise, and α is the fixed nominal level. The rule is simple, and the fixed target is the part that makes it work. The level moves toward the value at which the long-run error rate is α.

A common implementation error replaces the fixed α with the current α_t. The recursion becomes $\alpha_{t+1} = \alpha_t + \gamma(\alpha_t - \mathrm{err}_t)$. The difference looks small, and it is not. The correct rule has a fixed point at err = α. The incorrect rule has a fixed point at err = α_t, and because the procedure is calibrated at every level, that condition is satisfied, to first order, at every α_t. There is no restoring force toward the nominal level. The rule has a positive feedback term. When the set covers, α_t increases by $\gamma\alpha_t$, and when it misses, α_t decreases by $\gamma(1 - \alpha_t)$. At $\alpha_t = 0.10$ the downward step is nine times the upward step, so a run of misses drives the level to the lower clip, where the sets are as large as the calibration scores allow. The procedure then reports a high coverage because the sets are large, not because the level tracked the target.

The experiment compares the two rules on controlled drift streams. On digits, the stream has 450 points in nine blocks, with Gaussian noise increasing from 0 to 0.60. On CIFAR-10, it has 3,000 images in three blocks of 1,000, one block for each severity, across five corruption types. The calibration set is clean in both cases. The digits grid varies the step size from 0.005 to 0.2, the update rule, and the seed; the CIFAR-10 grid uses a single step size, 0.02. Table 4 and Table 7 report the results, and Figure 4 plots the digits coverage and level trajectories for the linear schedule at $\gamma = 0.02$.

On digits, the static procedure covers 0.666 of the stream. The correct update at $\gamma = 0.02$ covers 0.836 with a mean set size of 3.44. The incorrect update covers 0.869 with a mean set size of 4.58. The coverage gain is 3.3 points. The set-size cost is 33 percent. The share of near-trivial sets, those containing at least nine of the ten labels, rises from 10.8 to 24.4 percent. The final level is 0.0074 for the correct rule and 0.0011 for the incorrect one, which is the lower clip. On CIFAR-10 with five corruption types and three severities, the static procedure covers 0.451. The correct update covers 0.828 with a mean size of 4.93. The incorrect update covers 0.936 with a mean size of 6.97. The coverage gain is 10.8 points, the set-size cost is 41 percent, and the near-trivial share rises from 14.5 to 38.0 percent. In both image streams the incorrect rule wins the coverage comparison and loses the information comparison.

The correct rule also fails to reach 0.90 on these streams. On digits it reaches 0.836 with the level at 0.0074. On CIFAR-10 it reaches 0.828 with the level at 0.0106. The failure is not a bug in the update. It is the finite-time behavior of the method. The calibration scores bound how far the quantile can move. Once the level is at the clip, the

set is the largest the calibration sample allows, and a corrupted point can still fall outside it. The step size bounds how fast the level can move, and a drift that is faster than the step size produces a transient loss. The two mechanisms are visible in the block curves of Figure 4 for digits and in Table 7 for CIFAR-10. The correct rule tracks the target at the beginning of the stream and falls behind as the corruption increases.

The incorrect rule is also unstable. At $\gamma = 0.1$ on the abrupt schedule, two of the five digits seeds drive the level above 0.69, one of them to 0.998, and coverage falls to 0.436 and 0.162. The seed standard deviation of coverage is 0.205 on the linear schedule, against 0.046 for the correct rule at the same step size. The positive feedback term can push the level to either clip, and the direction depends on the sequence of errors. A method that reports a single coverage number and hides the level trajectory can present a divergent run as a successful one.

The library audit in Section 4.4 changes the scope of this result. MAPIE, TorchCP, and AdaptiveConformal all use the fixed-target update. The error is a risk of hand-written implementations, not a default of the libraries we checked. The audit's value is therefore narrower than "the ecosystem is wrong." It is that the error is easy to make, it produces a plausible-looking improvement, and a coverage report alone cannot detect it. The report needs the level trajectory, the clip fraction, and the set-size distribution. Section 4.4 gives the library evidence, and Section 6 turns it into a checklist.

4.4 Do real libraries do this?

The audit so far uses our own set constructor. A reader may ask whether the behaviour is an artifact of that choice. We inspected eight implementations: MAPIE 1.6.0, TorchCP 1.2.1, puncc 0.9.3, crepes 0.9.1, coverforest 0.0.3, nonconformist 2.1.0, conformal 0.1, and the AdaptiveConformal R package. All eight were read at the source level. MAPIE, puncc, crepes, nonconformist, and conformal were also run on a common synthetic five-class problem; MAPIE was run with five seeds, and the others with one.

The first finding is that the empty-set policy is a design choice, not a universal. MAPIE's APS implementation defaults to include_last_label=True, which adds the label that crosses the quantile and therefore never returns an empty set. TorchCP returns the boolean mask scores <= q_hat. puncc returns index lists. crepes returns p_values >= 1 - confidence. All three can return empty sets. coverforest exposes the choice directly, with allow_empty_sets=True as the default and documentation that recommends using False sparingly. Table 8 summarises the eight implementations. It is a policy map, not a defect list. The sample includes two older, smaller packages as well as the maintained ones. It is a sample of public implementations, not a census.

The second finding is that MAPIE's default is a non-empty rule of a different kind from the argmax fallback studied in Section 4.2. It does not insert the argmax. It inserts the last label whose cumulative mass crosses the threshold. The effect is nevertheless the same in kind. Over five seeds at alpha = 0.10, the default reaches 0.9457 +/- 0.0107 with zero empty sets and a mean size of 3.676. The randomized setting, which is the exact conformal rule, reaches 0.8998 +/- 0.0105 with an empty rate of 0.0004 and a mean size of 3.232. The False setting reaches 0.8284 +/- 0.0150, below the target. The two non-randomized settings therefore err in opposite directions. The default over-covers by 4.6 points; the exact rule lands on the target.

One more library reproduces the pattern. On the same one-seed task, puncc's APS with randomization off reaches 0.977 with no empty sets. With randomization on it reaches 0.889 with an empty rate below 0.2 percent. The switch is explicit in the API, and the documentation says that turning randomization off avoids empty sets. The LAC implementations of MAPIE, puncc, and crepes agree to three decimals (0.895, mean size 2.87), which is a consistency check on the audit itself.

The MAPIE documentation acknowledges that True and False "may result in a coverage higher than 1 - alpha", because neither creates empty sets. It does not quantify the gap, and it does not say that the default differs from the exact rule by enough to change a comparison between methods. That is the gap this section fills.

This matters for how the result should be read. The phenomenon in Section 4.2 is not specific to our argmax fallback. Any non-empty rule changes the coverage of the sets it produces, and the direction and size of the change

depend on the rule. The argmax fallback produces the identity derived in Section 4.2, where the gain is the empty rate times the conditional accuracy of the argmax. The MAPIE default produces a different accounting, but the same qualitative outcome: coverage above the nominal level, larger sets, and a reported number that belongs to the post-processing rather than to the guarantee.

The third finding concerns the ACI update. MAPIE's time-series regressor updates the level with a fixed target, alpha_t + gamma * (alpha_0 - error). TorchCP does the same with a fixed alpha. The AdaptiveConformal R package updates a quantile level against a fixed alpha. None of the three uses the alpha_t recursion studied in Section 4.3. The error is therefore a risk of hand-written implementations, not a default of the libraries we checked. That narrows the claim of Section 4.3. The value of the audit is not that the libraries are wrong. It is that the error is easy to make, produces a plausible-looking improvement, and is invisible in a coverage report alone.

Two practical consequences follow. A user of MAPIE's default APS should report 0.946, not 0.90, and should say that the number comes from a non-empty rule. A user who wants the exact guarantee should select the randomized setting, or report both the unmodified sets and the non-empty version. More generally, a coverage report is not interpretable without the set-construction rule that produced it. The rule is part of the method, and it belongs in the methods section.

The audit has limits. Five of the eight implementations were run. TorchCP, coverforest, and AdaptiveConformal were read at the source level. coverforest could not be run against scikit-learn 1.9, and TorchCP's optional graph dependencies are not available on this machine. The versions are those available on 2026-11-06, and line numbers will move. The MAPIE experiment uses synthetic data, so the 4.6-point gap should be read as an illustration of the mechanism rather than an estimate for a particular application. The claim we defend is narrow: in the implementations we audited, the non-empty policy varies, one widely used default over-covers by 4.6 points, and the three ACI implementations we checked use the correct fixed-target update.

5. Experiments

5.1 Design

The experiments test three claims, not the accuracy of conformal prediction on a benchmark. The first claim is that a probability threshold is a calibration rule, and that the correct calibration is split conformal. The second is that non-empty post-processing changes the coverage a pipeline reports. The third is that the α_t update is not ACI. The design varies the quantities that should matter for these claims: the score, the target level, the calibration size, the model strength, and the dataset.

Three datasets are used: two image benchmarks and one tabular benchmark. The digits benchmark has 1,797 images of 8 by 8 pixels in ten classes. A stratified split holds out 450 test points and 898 training points, and the calibration set is drawn from the remaining 449. The model is a standardiser followed by multinomial logistic regression. The training set is subsampled to 5, 25, and 100 percent of the pool, which gives 44, 224, and 898 training points and a test accuracy that ranges from about 0.83 to 0.96. CIFAR-10 has 50,000 training images and 10,000 test images in ten classes. The images are passed through a pretrained ResNet-18, and the 512-dimensional penultimate features are used with a logistic-regression head. The training pool is 30,000 images, and the head is fitted on 100, 1,000, 5,000, or 30,000 of them, which gives a test accuracy from 0.69 to 0.86. The calibration set is drawn from the remaining 20,000, with sizes 500, 2,000, and 10,000. The official test set is used unchanged. Covertype is a 581,012-row, 54-feature, seven-class tabular dataset from the UCI repository [10]. We use a 120,000-row stratified subset, a logistic regression, training sizes of 6,000 and 60,000, a calibration set of 20,000, and a test set of 40,000. The audit B grid is repeated on it.

The benchmarks are chosen for contrast, not for scale. Digits is small enough that every number can be checked by hand, and CIFAR-10 is large enough that the coverage estimates are tight. None is a deployment benchmark, and Section 5.6 is explicit about the limits.

Four scores are compared. The naive score is $1 - p_y$. The randomized APS score accumulates the probability mass of the classes ranked above the true class and adds a uniform fraction of the true class's probability. The deterministic APS score uses the cumulative mass through the true class and no randomization. The RAPS variant adds a penalty on the number of classes in the set, with $\lambda = 0.01$ and $k_{reg} = 2$ fixed rather than tuned. The scores are implemented from the papers rather than imported from a library, so the audit measures the method, not a library default. Section 4.4 audits the libraries separately.

The post-processing is the variable that the audit turns on. Every cell is evaluated twice. The raw evaluation keeps empty sets as empty. The forced evaluation inserts the argmax into every empty set. The forced evaluation is a measurement device, not a method. It answers the question "what would this pipeline report if it forced a non-empty output?" The two evaluations share the same sets, the same scores, and the same calibration. The only difference is the post-processing.

The metrics are coverage, mean set size, empty rate, class-conditional coverage, expected calibration error, and negative log-likelihood. Coverage and size are the primary quantities. The empty rate is the mechanism behind the post-processing result. The class-conditional coverage is reported because it is often confused with pointwise conditional coverage; Section 2.4 explains the difference. The calibration metrics are reported because the scores are built on probability outputs, and a reader should be able to see whether the model is over- or under-confident.

Every cell is repeated over five seeds, except the CIFAR-10 drift experiment, which uses three. The digits confidence intervals are bootstrap intervals over the test points; the CIFAR-10 intervals are normal approximations, which are tight at 10,000 test points. The seeds change the split, the calibration draw, and the model fit. The reported numbers are means over seeds, and the tables report the standard deviation where it matters.

Table 10 summarises the grid. The full grid is larger than the tables. The digits audit B has 540 cells and the CIFAR-10 audit B has 720. The tables report the cells at $\alpha = 0.10$, and the figures report the full range.

5.2 Audit A: the threshold is a calibration rule

Audit A compares four threshold rules at $\alpha = 0.10$. The rules are the raw threshold at 0.90, the max-probability quantile, the true-class quantile, and split conformal. They are evaluated on digits with $n_{cal} = 449$ and on CIFAR-10 with $n_{cal} = 2{,}000$, across the model strengths.

The result is a check, not a discovery. The true-class rule and split conformal are identical pointwise. The raw threshold is degenerate. The max-probability rule is between the two and is not exact. Table 2 and Table 5 give the numbers, Figure 2 plots them, and Section 4.1 reads the table. The purpose of the audit is to establish the baseline for the next two. If the threshold is a calibration rule, then the interesting comparison is between scores and between post-processing rules, not between conformal and a raw threshold.

5.3 Audit B: the post-processing

Audit B varies the score, the target level, the calibration size, and the model strength. On digits the grid has 540 cells. On CIFAR-10 it has 720. Each cell is evaluated raw and forced, so the audit produces 1,080 and 1,440 rows. The full results are in Table 3 and Table 6, and Figure 3 plots the change against the empty rate.

The summary is in Section 4.2. Three features of the design are worth stating here. First, the grid is balanced over the score and the calibration size, so the ranking of the scores is not an artifact of one calibration size. Second, the model strength is varied independently of the calibration size, because the training pool and the calibration pool are disjoint and the calibration size is drawn from a fixed pool. Third, the forced evaluation is applied to the same sets as the raw evaluation, so the change is not confounded by a different threshold or a different score.

The grid also produces the exception that Section 4.2 discusses. Deterministic APS at 5,000 CIFAR-10 images is already conservative before the fallback, with a coverage of 0.994 and a mean set size of 6.08. The fallback has nothing to do. The cell is not a failure of the audit. It is a reminder that over-coverage can come from the score as well as from the post-processing.

Table 11 repeats the audit on covertype. At alpha = 0.10 the empty rate is 0.0 to 0.4 percent, and the forcing change is 0.0 to 0.4 points. The identity holds; the magnitude is governed by the empty rate. The trap is a regime effect, not a universal constant: on a well-calibrated tabular model with seven classes, the same post-processing is nearly invisible. The near-zero change is what the identity predicts when the empty rate is near zero; it is a confirmation, not a null result. The empty rate is the diagnostic, and it travels across datasets.

5.4 Audit C: the update rule

Audit C compares two update rules on controlled drift streams. The digits stream has 450 points in nine blocks, with Gaussian noise increasing from 0 to 0.60. Three schedules are used: linear, step, and abrupt. The CIFAR-10 stream has 3,000 images in three blocks of 1,000, one for each severity of a corruption type, across five corruption types. The calibration set is clean in both cases, so the drift is a violation of exchangeability rather than a corrupted calibration.

The digits grid varies the step size over six values from 0.005 to 0.2, the update rule over two values, and the seed over five. The CIFAR-10 grid uses a single step size, 0.02, five corruption types, and three seeds. The static procedure is included as a baseline. Table 4 and Table 7 report the results, and Figure 4 plots the coverage and the level trajectories for the linear schedule at $\gamma = 0.02$.

The design separates the two questions that a coverage number confounds. The first is whether the method adapts. The second is whether the adaptation is bought with larger sets. The grid reports both, along with the level trajectory, the fraction of steps at a clip bound, and the share of near-trivial sets. A reader who only looks at the coverage cannot answer either question.

5.5 The regime map

The 1,260 image cells of audit B, plus the covertype replication, are the raw material for a different view. Figure 5 plots the raw empty rate and the coverage gain against the test accuracy, at $\alpha = 0.10$ and a calibration size of 449 for digits, 500 for CIFAR-10, and 20,000 for covertype. Each point is a model strength, averaged over seeds. The figure compares the naive score and the randomized APS. Covertype sits near the origin: the empty rate is 0.0 to 0.4 percent, so the change is 0.0 to 0.4 points. The third dataset is the low end of the diagnostic, not a counterexample to it.

The view is not a model-strength story. On digits the empty rate rises with accuracy, from 0 to 7.8 percent for the naive score and from 3.4 to 7.0 percent for APS. On CIFAR-10 the naive score is at zero at every strength, and APS is between 1.0 and 3.5 percent. The difference is the score and the calibration, not the accuracy. A stronger model can have a higher empty rate if the score is more likely to cross the threshold on every label. The figure makes the mechanism visible and prevents the reader from reading the audit as a statement about weak models.

The 5,000-image APS cell is the exception again. It is the only point where the empty rate falls below the neighboring strengths, and it is the point where the score is already conservative. The exception is a useful warning. The empty rate is not a monotone function of accuracy, and a reader who expects a smooth curve will misread the figure.

5.6 What the experiments do and do not show

The experiments show that the three mechanisms are real and that they are reproducible in direction across three datasets and a range of model strengths, though the magnitudes vary with the empty rate. They also show that on digits and CIFAR-10 the magnitudes are large enough to change a comparison. The raw threshold can turn 70 percent of a test set into abstentions and then report the model's accuracy. At $\alpha = 0.10$, the post-processing can add 9.7 points of coverage on digits and 9.5 points on CIFAR-10, with no change to the score. On covertype it adds 0.0 to 0.4 points, because the empty rate is 0.0 to 0.4 percent. The wrong update can add 10.8 points of coverage on CIFAR-10 while inflating the set size by 41 percent. On digits and CIFAR-10 these are not small effects, and they are not artifacts of a single seed. On covertype they are small, and the empty rate explains why.

The experiments do not show that the traps are common across all libraries. Section 4.4 shows the opposite for the update rule: three libraries use the fixed-target update. They do not show that the magnitudes transfer to ImageNet, to language models, or to a deployment with a different model class. The models are linear heads on small images and a tabular benchmark; the drift is a synthetic ramp on digits and a corruption family on CIFAR-10; the datasets are three. The results are about mechanisms, not about scale.

The experiments also do not establish that the correct ACI is a good method. On the streams we tested it fails to reach the target, for reasons that Section 4.3 explains. A reader who wants a positive result about ACI will not find it here. A reader who wants to know how the method fails, and how the failure can be mistaken for success, will.

The code is a single script that fixes the seeds and writes every reported number to a JSON file. The tables and figures are generated from that file. The data are public, and the experiments can be rerun on a laptop. The reproducibility is not a claim about the importance of the results. It is a condition for the reader to check them.

6. A checklist for practitioners

The audit suggests a short checklist. None of the items is new. What is new is how often they are omitted, and how much the reported number moves when they are.

1. Report the empty rate before any post-processing. The empty rate is the mechanism behind the post-processing result. A forced coverage number without the raw empty rate cannot be compared to an unforced one. Section 4.2 gives the identity, and the empty rate is its first factor.
2. Calibrate the threshold on the true-class probability, not on the maximum probability. The coverage statement is about the true label. The max-probability rule is exact only when the argmax is always correct, which is a property of the model, not of the procedure. Section 4.1 shows the gap on CIFAR-10, where the max-probability rule reaches 0.810 at a target of 0.90.
3. State the update rule and the target it tracks. For adaptive conformal inference, write the recursion and say whether the target is fixed. Section 4.3 shows that the two rules have different fixed points, and that the difference is invisible in a coverage report.
4. Report the level trajectory, the clip fraction, and the set-size distribution. Coverage alone cannot distinguish adaptation from inflation. The incorrect rule in Section 4.3 reports higher coverage with larger sets, and the trajectory makes the difference visible. A single number can present a divergent run as a success.
5. Match the post-processing rule before comparing methods. A forced method and an unforced method are two procedures, not two scores. The comparison should hold the post-processing fixed, and the empty rate should be reported for both.
6. Report the calibration size and the group definition. The finite-sample coverage is k/(n + 1) with continuous scores, and the calibration-conditional coverage is a random variable. A group-conditional claim is only as good as the group definition, and class-conditional coverage is not pointwise coverage.

6.1 When to force the argmax

The checklist says to report both numbers. It does not say which one to use. The choice is a decision under uncertainty, and it can be stated in one line.

Let c_a be the loss of abstaining on an empty set, and c_e the loss of a wrong answer. A correct answer has loss zero. If the set is empty and the procedure abstains, the loss is c_a. If it forces the argmax, the answer is correct with probability a_E, where a_E is the conditional accuracy of the argmax on the empty sets, and wrong with probability 1 - a_E. The expected loss of forcing is c_e (1 - a_E). Forcing is better than abstaining exactly when

c_e (1 - a_E) < c_a.

The rule is a threshold on the ratio c_a/c_e. If abstaining costs as much as an error, any positive a_E makes forcing worthwhile. If abstaining is free, forcing is never worthwhile. If abstaining costs half an error, forcing is worthwhile when a_E > 0.5. The audit does not choose the ratio. The deployment does.

The audit does supply a_E. The accounting identity of Section 4.2 gives Δ = empty rate × a_E, so a_E is the ratio of the coverage gain to the empty rate. A deployment that runs the audit on its own calibration and test data can estimate a_E, plug in its costs, and decide. The six numbers of the checklist are the inputs to the rule.

The library defaults are uniform versions of the decision, though not all of them insert the argmax. MAPIE's include_last_label=True adds the last label that crosses the quantile; puncc's rand=False adds labels until the set is non-empty. Both are non-empty policies of the same kind, and the same cost comparison applies with a_E replaced by the accuracy of the label the rule inserts. They are optimal for a deployment whose c_a is high relative to c_e (1 - a_E), and suboptimal for a deployment that values abstention. The library cannot know the ratio. This is why the default should be the exact rule, which carries the guarantee, and the non-empty rule should be an explicit option. coverforest exposes the option with allow_empty_sets. MAPIE and puncc expose it through include_last_label and rand, but their defaults choose the non-empty side. A default is a decision, and a statistical library is not the place to make a deployment's decision.

The rule can also be applied per point, with the model's maximum probability as a proxy for a_E. That version is a heuristic, not a guarantee. The conformal construction does not give the conditional accuracy of the argmax on empty sets, and the model's probability is not that quantity. The average rule is the one the audit supports. The per-point rule is a heuristic, and on the three datasets we tested it did not improve on the average rule (Section 6.2).

The conclusion is not that the fallback is wrong. It is that the fallback is a decision, and the decision should be made with the costs in view. The audit's contribution is to make the decision measurable. The empty rate, the conditional accuracy, and the coverage and size costs are the quantities the decision needs. The checklist reports them. The rule uses them.

6.2 A cost-sensitive reading

Table 12 applies the rule to the cells of audit B. For each cell with a non-zero empty rate, a_E is estimated as the coverage gain divided by the empty rate, and the threshold is 1 - a_E. The estimate is high in every dataset, between 0.94 and 0.99. The threshold is therefore small, between 0.01 and 0.06. Forcing is better than abstaining unless abstention costs less than one to six percent of an error.

This explains why the non-empty defaults are popular. They are not a mistake in every deployment. When the inserted label is right on most empty sets, forcing is the better bet, and the default saves the deployment from an arbitrary abstention. The problem is not that the bet is always wrong. The problem is that the bet is made silently: the default changes the guarantee, and the reported coverage no longer estimates the nominal level.

The empty rate determines the stakes. On digits, 6 to 9 percent of the points are affected, so the choice moves the reported coverage by several points. On covertype, 0.3 percent are affected, so the same choice moves it by less than half a point. The threshold is small in every dataset; the exposure is not.

We also tested the per-point version. On the same three datasets, at alpha = 0.10 with deterministic APS, it coincides with uniform forcing at or above c_a/c_e = 0.05: the empty sets are high-confidence, so the proxy forces them all. The per-point rule does not improve on the population rule. The negative result is consistent with the high a_E. When the argmax is right on 94 to 99 percent of the empty sets, there is little for a per-point rule to select on.

The practical recommendation is therefore narrower than "never force". It is: report the empty rate, report the conditional accuracy, and make the decision explicit. A deployment that prefers a non-empty output can force, and the cost analysis may justify it. A deployment that values abstention should not inherit that choice from a default.

A paper that reports these six quantities can be checked. A paper that reports only coverage cannot. The checklist is not a substitute for the theory. It is a condition for the theory to be visible in the reported number.

7. Open problems

The map in Table 1 has empty cells. They are the open problems, and they are stated here as measurable gaps rather than as declarations of impossibility. In each case, the question is what would count as progress, not whether the problem is hard.

7.1 Pointwise conditional coverage without exchangeability

The impossibility result of Barber et al. [7] is a statement about exchangeable data. Under covariate shift or non-exchangeability, the target is harder, and the map has no entry. Local weighting [22] obtains a conditional guarantee up to a smoothness constant, and the bound under non-exchangeability [8] is stated in terms of an unobservable distance. A procedure that gives a pointwise guarantee under a quantifiable departure from exchangeability, or a lower bound that shows the cost of the departure, would be progress. A procedure that assumes the departure away is not progress.

7.2 Group-conditional coverage under shift

The group-conditional column is empty under covariate shift and under non-exchangeability. The known-group case is solved by Mondrian calibration [37], and the weaker group-level guarantee is developed by Gibbs et al. [21]. The open case is the one in which the group label is missing at test time, or the group distribution shifts with the covariates. A group-weighted version of the covariate-shift correction [35] would be a natural candidate, but the weight estimation error and the group sample size interact, and the interaction is not understood. A bound on the group coverage under a known shift, or a procedure that is robust to a misspecified group model, would be progress.

7.3 Risk control under covariate shift

The risk-control row is empty under covariate shift in the reviewed set. The exchangeable construction [9] and its grid and cross-validated variants [4, 13] assume that the calibration and test distributions coincide. The natural extension is to weight the risk estimate by the likelihood ratio, but the weight estimation error enters the risk bound in a way that is not yet characterized, and the finite-sample correction for the weighted empirical risk is not settled. A risk bound that accounts for the weight estimation error, or a procedure that remains valid under a misspecified ratio, would be progress.

7.4 The finite-time behavior of adaptive inference

The correct update in Section 4.3 fails to reach the target on the streams we tested, for two reasons that are not separated by the current theory. The calibration support bounds how far the quantile can move, and the step size bounds how fast it can move. A bound on the realized coverage as a function of the stream length, the step size, and the calibration support would make the two mechanisms separable. A step-size schedule that closes the gap, or a diagnostic that distinguishes a support-limited failure from a step-limited one, would be progress. The current practice of reporting a single average coverage number does not distinguish them.

7.5 The prevalence and the utility of non-empty rules

Section 4.4 audits eight implementations. The policy map is a start, not a census. A systematic audit of the defaults across the ecosystem, including the libraries we did not run, would establish whether the MAPIE default is an outlier or a pattern. The utility question is separate. The audit measures the cost of a non-empty rule in coverage and set size, but it does not model the decision that the rule supports. A deployment that prefers a guess to an abstention may be better off with the fallback, and a deployment that acts on the set may be worse off. A utility model that maps the empty rate and the conditional accuracy to a decision, and an audit of the default policies against that model, would turn the checklist into a design rule.

The empty cells are not a criticism of the field. They are the places where the next paper can be useful. The audit's contribution is to make them visible, not to fill them.

Data and code availability

The experiments are implemented in a single Python repository with fixed seeds. Every empirical number is written to a JSON file, and the tables and figures are generated from those files. The digits benchmark is distributed with scikit-learn; CIFAR-10 and covertype are public. README.md documents the environment, the data, and the commands that regenerate Tables 2-7, 11 and 12 and Figures 1-5. Table 8's source-policy columns are a curated reading of the code; its runtime rows come from results/library_runtime.json. The repository is available at https://github.com/ZhangShaoXuan-HIT/conformal-audit and archived at https://doi.org/10.5281/zenodo.24001054.

Tables

Table 1. Guarantee by data assumption

Each cell gives a representative method, the assumption it needs, and the price it pays. “Open” means no method in the reviewed set covers that cell. The empty cells are the open problems of Section 7, not gaps in the table.

guarantee	exchangeable	covariate shift	non-exchangeable / temporal	known groups
marginal coverage	[38], [24], [3]. One calibration pass.	[35]. Likelihood ratio known or estimable.	[8] (Barber et al. 2023), [28]. Gap bounded by the violation.	[37] (Mondrian). Per-group quantiles; small n per group.
group-conditional coverage	[37], [21]. Group-wise quantiles; smaller effective n.	open	open	[37], [21]. Group labels known at test.
pointwise conditional coverage	[7] (impossible without assumptions), [22] (local weights and smoothness). Larger sets.	open	open	open (implied by pointwise)
risk control	[9], [5] (Angelopoulos et al. 2022), [4] (Angelopoulos et al. 2025). Monotone loss; grid and multiplicity.	open in the reviewed set	[17] (Farinhas et al. 2023). Bounded non-exchangeability.	[14] (Macro Mondrian [5]).

Table 2. A calibrated threshold is split conformal

digits, alpha = 0.10, n_cal = 449, 5 seeds. Coverage is the mean over seeds. Raw keeps empty sets; forced fills them with the argmax. Size is mean cardinality. Pointwise set equality between the conformal threshold and split conformal: mismatches = 0.

strength = 0.05 (n_train = 44)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.294	0.833	0.299	1.000	0.701
threshold at max-prob quantile	0.800	0.837	0.920	1.006	0.086
threshold at true-class quantile	0.918	0.918	1.389	1.390	0.000
threshold at t = 1-q (conformal)	0.919	0.919	1.396	1.397	0.000
split conformal, naive score	0.919	0.919	1.396	1.397	0.000
split conformal, APS	0.917	0.952	1.886	1.922	0.036

strength = 0.25 (n_train = 224)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.713	0.938	0.719	1.000	0.281
threshold at max-prob quantile	0.886	0.938	0.912	1.000	0.088
threshold at true-class quantile	0.909	0.939	0.949	1.002	0.053
threshold at t = 1-q (conformal)	0.911	0.939	0.951	1.002	0.051
split conformal, naive score	0.911	0.939	0.951	1.002	0.051
split conformal, APS	0.904	0.976	1.178	1.251	0.073

strength = 1.00 (n_train = 898)

method	coverage raw	coverage forced	size raw	size forced	empty raw
softmax threshold, raw t = 1-alpha	0.855	0.964	0.860	1.000	0.140

method	coverage raw	coverage forced	size raw	size forced	empty raw
threshold at max-prob quantile	0.901	0.964	0.912	1.000	0.088
threshold at true-class quantile	0.908	0.964	0.920	1.000	0.080
threshold at t = 1-q (conformal)	0.910	0.964	0.922	1.000	0.078
split conformal, naive score	0.910	0.964	0.922	1.000	0.078
split conformal, APS	0.903	0.982	1.034	1.113	0.079

Table 3. Non-empty post-processing inflates coverage

digits, alpha = 0.10, 5 seeds. Delta = forced coverage - raw coverage. Raw empty is the share of sets that were empty before the argmax was forced in.

strength = 0.05 (n_train = 44)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.932	0.932	+0.000	0.000	1.565	1.565
naive	225	0.929	0.929	+0.000	0.000	1.491	1.491
naive	449	0.919	0.919	+0.000	0.000	1.396	1.397
APS randomized	100	0.914	0.954	+0.040	0.040	1.983	2.024
APS randomized	225	0.918	0.952	+0.033	0.034	1.864	1.898
APS randomized	449	0.920	0.954	+0.033	0.034	1.922	1.956
APS deterministic	100	0.893	0.981	+0.088	0.090	3.551	3.641
APS deterministic	225	0.899	0.984	+0.085	0.086	3.619	3.705
APS deterministic	449	0.898	0.981	+0.084	0.084	3.549	3.634
RAPS	100	0.922	0.952	+0.030	0.031	1.941	1.972
RAPS	225	0.917	0.951	+0.034	0.034	1.874	1.908
RAPS	449	0.911	0.945	+0.034	0.034	1.781	1.816

strength = 0.25 (n_train = 224)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.922	0.947	+0.025	0.044	0.981	1.025
naive	225	0.912	0.941	+0.029	0.050	0.956	1.007
naive	449	0.911	0.939	+0.028	0.051	0.951	1.002
APS randomized	100	0.911	0.979	+0.068	0.068	1.236	1.305
APS randomized	225	0.901	0.975	+0.074	0.074	1.193	1.268
APS randomized	449	0.919	0.978	+0.059	0.060	1.239	1.299
APS deterministic	100	0.892	0.995	+0.104	0.105	3.220	3.325
APS deterministic	225	0.907	0.996	+0.089	0.090	3.372	3.463
APS deterministic	449	0.898	0.995	+0.097	0.099	3.266	3.364
RAPS	100	0.910	0.975	+0.065	0.065	1.215	1.280
RAPS	225	0.894	0.973	+0.080	0.080	1.186	1.266
RAPS	449	0.896	0.973	+0.077	0.077	1.148	1.226

strength = 1.00 (n_train = 898)

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
naive	100	0.929	0.964	+0.035	0.056	0.944	1.000
naive	225	0.917	0.964	+0.047	0.071	0.929	1.000
naive	449	0.910	0.964	+0.054	0.078	0.922	1.000
APS randomized	100	0.903	0.984	+0.081	0.081	1.049	1.130
APS randomized	225	0.899	0.981	+0.082	0.082	1.044	1.126
APS randomized	449	0.913	0.983	+0.070	0.070	1.068	1.137

score	n_cal	coverage raw	coverage forced	delta	empty raw	size raw	size forced
APS deterministic	100	0.894	1.000	+0.106	0.106	3.011	3.117
APS deterministic	225	0.908	1.000	+0.092	0.092	3.127	3.218
APS deterministic	449	0.911	1.000	+0.089	0.089	3.110	3.199
RAPS	100	0.900	0.982	+0.081	0.082	1.028	1.110
RAPS	225	0.887	0.982	+0.095	0.096	1.018	1.113
RAPS	449	0.884	0.980	+0.096	0.096	1.001	1.098

Table 4. The alpha_t update is not ACI

digits, alpha = 0.10, 5 seeds. Coverage and size are means over seeds. Min/max block are computed on the seed-averaged block curve. Clip is the share of steps at a clip bound.

schedule = linear

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.666	0.100	1.524	0.388	0.968	0.1000	0.000	0.000
fixed	0.005	0.832	0.023	3.371	0.776	0.948	0.0054	0.045	0.130
fixed	0.010	0.840	0.030	3.598	0.768	0.932	0.0064	0.071	0.139
fixed	0.020	0.836	0.038	3.444	0.732	0.928	0.0074	0.091	0.108
fixed	0.050	0.818	0.049	3.469	0.684	0.916	0.0090	0.132	0.105
fixed	0.100	0.804	0.046	3.182	0.668	0.908	0.0090	0.156	0.087
fixed	0.200	0.775	0.043	3.103	0.624	0.904	0.0170	0.203	0.098
buggy	0.005	0.849	0.025	3.773	0.784	0.940	0.0010	0.052	0.176
buggy	0.010	0.865	0.031	4.273	0.780	0.924	0.0010	0.061	0.216
buggy	0.020	0.869	0.036	4.580	0.752	0.944	0.0011	0.065	0.244
buggy	0.050	0.870	0.043	5.169	0.792	0.956	0.0012	0.070	0.297
buggy	0.100	0.781	0.205	4.868	0.668	0.868	0.0014	0.076	0.279
buggy	0.200	0.769	0.153	4.788	0.476	0.900	0.0019	0.091	0.260

schedule = step

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.663	0.085	1.488	0.432	0.960	0.1000	0.000	0.000
fixed	0.005	0.830	0.024	3.358	0.576	0.948	0.0064	0.051	0.130
fixed	0.010	0.835	0.028	3.492	0.680	0.940	0.0102	0.077	0.132
fixed	0.020	0.828	0.039	3.349	0.732	0.928	0.0098	0.102	0.104
fixed	0.050	0.808	0.046	3.176	0.712	0.916	0.0120	0.138	0.088
fixed	0.100	0.790	0.047	3.013	0.684	0.916	0.0190	0.166	0.079
fixed	0.200	0.767	0.042	3.010	0.664	0.900	0.0170	0.208	0.084
buggy	0.005	0.836	0.028	3.575	0.564	0.944	0.0010	0.060	0.161
buggy	0.010	0.845	0.031	3.826	0.676	0.924	0.0011	0.072	0.178
buggy	0.020	0.839	0.040	3.928	0.712	0.924	0.0012	0.076	0.188
buggy	0.050	0.809	0.067	4.078	0.636	0.892	0.0016	0.078	0.202
buggy	0.100	0.748	0.183	4.042	0.620	0.852	0.0015	0.086	0.199
buggy	0.200	0.733	0.163	4.262	0.484	0.840	0.0024	0.092	0.214

schedule = abrupt

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
static	—	0.739	0.042	1.453	0.360	0.960	0.1000	0.000	0.000
fixed	0.005	0.831	0.021	2.411	0.508	0.948	0.0054	0.040	0.070
fixed	0.010	0.840	0.023	2.574	0.608	0.940	0.0064	0.061	0.073
fixed	0.020	0.836	0.029	2.499	0.640	0.928	0.0074	0.080	0.062
fixed	0.050	0.824	0.033	2.425	0.644	0.916	0.0090	0.103	0.052
fixed	0.100	0.816	0.030	2.398	0.620	0.916	0.0090	0.124	0.048

update	gamma	coverage	seed SD	size	min block	max block	alpha final	clip	sets >= 9
fixed	0.200	0.796	0.022	2.528	0.612	0.900	0.0170	0.167	0.061
buggy	0.005	0.828	0.024	2.448	0.492	0.944	0.0010	0.042	0.079
buggy	0.010	0.832	0.039	2.598	0.588	0.924	0.0010	0.052	0.085
buggy	0.020	0.807	0.065	2.580	0.564	0.924	0.0011	0.054	0.089
buggy	0.050	0.767	0.107	2.904	0.636	0.892	0.0012	0.062	0.115
buggy	0.100	0.628	0.320	2.500	0.492	0.852	0.3387	0.048	0.092
buggy	0.200	0.576	0.259	2.649	0.316	0.776	0.0019	0.066	0.122

CIFAR-10, audit A: a calibrated threshold is split conformal

alpha = 0.10, n_cal = 2000, 5 seeds, 10k official test set. Coverage is the mean over seeds. Raw keeps empty sets; forced fills them with the argmax.

n_train	method	coverage raw	coverage forced	size raw	empty raw
100	raw threshold 0.90	0.290	0.692	0.310	0.690
100	max-prob quantile	0.668	0.702	0.921	0.103
100	true-class quantile	0.898	0.898	2.086	0.000
100	t = 1-q	0.898	0.898	2.090	0.000
100	split conformal	0.898	0.898	2.090	0.000
100	APS	0.896	0.911	2.471	0.016
100	**accuracy**	**0.692**			
1000	raw threshold 0.90	0.647	0.803	0.705	0.295
1000	max-prob quantile	0.762	0.803	0.899	0.101
1000	true-class quantile	0.894	0.894	1.344	0.000
1000	t = 1-q	0.894	0.894	1.346	0.000
1000	split conformal	0.894	0.894	1.346	0.000
1000	APS	0.892	0.911	1.497	0.021
1000	**accuracy**	**0.803**			
5000	raw threshold 0.90	0.731	0.818	0.819	0.181
5000	max-prob quantile	0.771	0.818	0.892	0.108
5000	true-class quantile	0.899	0.899	1.286	0.000
5000	t = 1-q	0.899	0.899	1.287	0.000
5000	split conformal	0.899	0.899	1.287	0.000
5000	APS	0.897	0.907	1.351	0.011
5000	**accuracy**	**0.818**			
30000	raw threshold 0.90	0.716	0.855	0.756	0.244
30000	max-prob quantile	0.810	0.855	0.899	0.101
30000	true-class quantile	0.888	0.888	1.089	0.001
30000	t = 1-q	0.888	0.888	1.090	0.000
30000	split conformal	0.888	0.888	1.090	0.000
30000	APS	0.887	0.927	1.258	0.041
30000	**accuracy**	**0.855**			

CIFAR-10, audit B: non-empty post-processing inflates coverage

alpha = 0.10, n_cal = 2000, 5 seeds, 10k official test set. Delta = forced coverage - raw coverage. Raw empty is the share of empty sets.

n_train	score	coverage raw	coverage forced	delta	empty raw	size raw	size forced
100	naive	0.898	0.898	+0.000	0.000	2.090	2.090
100	APS randomized	0.898	0.913	+0.015	0.015	2.482	2.497
100	APS deterministic	0.904	0.972	+0.068	0.069	4.437	4.506
100	RAPS	0.899	0.911	+0.012	0.013	2.403	2.415
1000	naive	0.894	0.894	+0.000	0.000	1.346	1.346
1000	APS randomized	0.896	0.914	+0.018	0.019	1.515	1.534

n_train	score	coverage raw	coverage forced	delta	empty raw	size raw	size forced
1000	APS deterministic	0.902	0.993	+0.091	0.091	4.654	4.745
1000	RAPS	0.897	0.913	+0.016	0.016	1.490	1.506
5000	naive	0.899	0.899	+0.000	0.000	1.287	1.287
5000	APS randomized	0.900	0.909	+0.009	0.010	1.362	1.372
5000	APS deterministic	0.994	0.994	+0.000	0.000	6.083	6.083
5000	RAPS	0.899	0.907	+0.008	0.008	1.338	1.346
30000	naive	0.888	0.888	+0.000	0.000	1.090	1.091
30000	APS randomized	0.893	0.930	+0.036	0.037	1.271	1.309
30000	APS deterministic	0.899	0.994	+0.095	0.095	3.967	4.061
30000	RAPS	0.894	0.929	+0.035	0.036	1.261	1.297

Table 7. CIFAR-10-C style drift

Five corruption types (gaussian_noise, defocus_blur, contrast, brightness, jpeg_compression), three severities (1, 3, 5), 1,000 images per block, 3,000 points per stream, 3 seeds, alpha = 0.10, gamma = 0.02. Calibration is clean.

update	coverage	mean size	block sev 1	block sev 3	block sev 5	alpha final	clip	sets >= 9
static	0.451	1.30	0.588	0.457	0.308	—	—	0.000
fixed	0.828	4.93	0.853	0.829	0.802	0.0106	0.13	0.145
buggy	0.936	6.97	0.940	0.939	0.928	0.0019	0.05	0.380

Per corruption

Coverage and mean set size, averaged over three seeds.

corruption	static	fixed	buggy
gaussian_noise	0.22 / 1.34	0.66 / 5.45	0.81 / 7.02
defocus_blur	0.24 / 1.37	0.85 / 7.52	0.98 / 9.42
contrast	0.52 / 1.33	0.87 / 4.94	0.98 / 7.42
brightness	0.80 / 1.15	0.90 / 1.88	0.94 / 3.70
jpeg_compression	0.47 / 1.32	0.86 / 4.87	0.97 / 7.28

The incorrect update reports 10.8 points more coverage than the correct one, at a 41 percent larger mean set size, and raises the share of near-trivial sets from 14.5 to 38.0 percent. The correct update also falls short of 0.90, at 0.828.

Table 8. Library behavior audit

Versions checked on 2026-11-06. Line numbers refer to the named files in the package versions listed; the source snapshots were downloaded locally for the audit. Runtime: MAPIE, puncc, crepes, nonconformist, and conformal. Source inspection: TorchCP, coverforest, and AdaptiveConformal.

library	version	task	empty sets possible	default policy	evidence	ACI update
MAPIE	1.6.0	classification	LAC yes; APS policy-dependent	APS default include_last_label=True: no empty sets, coverage 0.9457 vs 0.90	classification.py:238-283; aps.py:355-358; lac.py:205-207; results/mapie_empty.json	fixed alpha_0 (time_series_regression.py:329)
TorchCP	1.2.1	classification	yes	no forcing; boolean mask (scores <= q_hat)	classification/predictor/base.py:111	fixed alpha (regression/predictor/aci.py:197)
puncc	0.9.3	classification	yes	no forcing; index lists, empty when no class passes	api/prediction_sets.py: 70-78, 112-119	not implemented
crepes	0.9.1	classification	yes	no forcing; boolean mask (p_values >= 1-confidence)	base.py:340	not implemented
nonconformist	2.1.0	classification	yes	no forcing; boolean matrix of p-values > significance; smoothing=True	icp.py:199-250	not implemented

library	version	task	empty sets possible	default policy	evidence	ACI update
conformal	0.1	classification	yes (empty list possible)	no forcing; per-label thresholds at 100 - epsilon percentile	**init**.py:76-93, 95-108	not implemented
coverforest	0.0.3	classification	yes by default	allow_empty_sets=True default; False avoids empty sets; docs warn to use it sparingly	_forest.py:1457, 2006-2008	not applicable
AdaptiveConformal	GitHub main	regression intervals	not applicable	not applicable	R/aci_algorithm.R:78-81	fixed alpha (correct)

MAPIE 1.6.0, five seeds, alpha = 0.10

Synthetic five-class data, logistic regression, 2,000 test points per seed. The correct randomised APS is the last row. The default is the second row.

score	include_last_label	coverage (mean +/- SD)	empty rate	size
LAC	True	0.9016 +/- 0.0080	0.0000	3.151
APS, default	True	0.9457 +/- 0.0107	0.0000	3.676
APS	False	0.8284 +/- 0.0150	0.0000	2.684
APS, randomized	randomized	0.8998 +/- 0.0105	0.0004	3.232

The default APS setting over-covers by 4.6 points with zero empty sets. The randomised setting, which is the exact conformal rule, lands on the target. The difference is the non-empty post-processing, not the score.

Runtime cross-check

Same synthetic five-class task for all libraries: 8,000 samples, logistic regression, 2,000 test points, alpha = 0.10, test accuracy 0.487. One seed for every library in this table; the MAPIE numbers in the table above use five seeds. The puncc randomized row has no fixed seed; its empty rate was 0.0 to 0.2 percent across runs.

library	setting	coverage	empty rate	mean size
MAPIE	LAC	0.895	0.000	2.87
MAPIE	APS, include_last_label=True (default)	0.949	0.000	3.46
MAPIE	APS, include_last_label=False	0.809	0.000	2.48
MAPIE	APS, randomized	0.892	0.001	3.02
puncc	LAC	0.895	0.000	2.87
puncc	APS, rand=True	0.889	0.000	2.99
puncc	APS, rand=False	0.977	0.000	4.06
crepes	LAC (hinge), p >= alpha	0.895	0.000	2.87
nonconformist	ICP margin, smoothing=True	0.890	0.000	3.50
conformal	per-label thresholds, epsilon=10	0.890	0.000	3.02

coverforest 0.0.3 could not be run on scikit-learn 1.9: its forest code calls _get_n_samples_bootstrap without the sample_weight argument that the installed scikit-learn requires. TorchCP 1.2.1 was not run: its optional graph dependencies (torch_geometric) are not installed. Both were read at the source level.

Table 9. Coverage against four surveys

Authors' assessment of topical coverage, not of quality. A circle means the topic is not a focus of that survey, not that the survey is deficient. Symbols: full circle = comprehensive, half circle = partial, open circle = not a focus.

dimension	this paper	Angelopoulos & Bates 2021	Fontana et al. 2023	Campos et al. 2024	Zhou et al. 2025
foundations	half	full	full	half	full
scores	full	full	half	full	full
conditional coverage	full	half	full	half	half
risk control	full	half	half	half	half

dimension	this paper	Angelopoulos & Bates 2021	Fontana et al. 2023	Campos et al. 2024	Zhou et al. 2025
non-exchangeable	full	open	half	half	half
language models	half	open	open	full	half
implementation audit	full	open	open	open	open
two-dataset experiments	full	open	open	open	open

The last two rows are the reason this paper exists. The first six rows are the reason it has to cite the other four.

Table 10. Experimental grid

dataset	train pool	strengths	calibration	test	model	scores	alpha	seeds
digits	898	44 / 224 / 898	449 for A and C; 100 / 225 / 449 for B	450	standardiser + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5
CIFAR-10	30,000	100 / 1,000 / 5,000 / 30,000	2,000 for A; 500 / 2,000 / 10,000 for B; 2,000 for C	10,000	ResNet-18 features + logistic regression	naive, APS randomized, APS deterministic, RAPS	0.10 for A and C; 0.05 / 0.10 / 0.20 for B	5 for A and B; 3 for C
covertype	120,000-row subset	6,000 / 60,000	20,000	40,000	standardiser + logistic regression	naive, APS randomized, APS deterministic	0.05 / 0.10 / 0.20	5

The calibration column gives the sizes used in each audit. The strength column gives the number of training points used to fit the model; the calibration and test sets are disjoint from it.

Table 11. Covertype audit B

120,000-row stratified subset, 54 features, 7 classes, logistic regression, 20,000 calibration points, 40,000 test points, alpha = 0.10, 5 seeds.

train	score	raw coverage	raw empty	raw size	forced coverage	forced size	delta
6000	naive	0.900	0.000	1.53	0.900	1.53	0.000
6000	APS randomized	0.901	0.003	1.67	0.904	1.68	0.003
6000	APS deterministic	0.900	0.004	1.92	0.903	1.92	0.004
60000	naive	0.901	0.000	1.51	0.901	1.51	0.000
60000	APS randomized	0.900	0.003	1.65	0.903	1.66	0.003
60000	APS deterministic	0.899	0.001	1.89	0.900	1.89	0.001

The two training sizes give test accuracies of 0.717 and 0.721.

delta is forced coverage minus raw coverage. The identity delta = empty rate x conditional accuracy holds. The empty rate is 0.0 to 0.4 percent, so the change is 0.0 to 0.4 points. The trap is a regime effect, not a universal constant.

Table 12. Cost-sensitive reading of the fallback

At alpha = 0.10, over cells with a non-zero raw empty rate. a_E is the conditional accuracy of the inserted label on empty sets, estimated as delta / empty rate. The threshold is the cost ratio c_a / c_e above which forcing is better than abstaining.

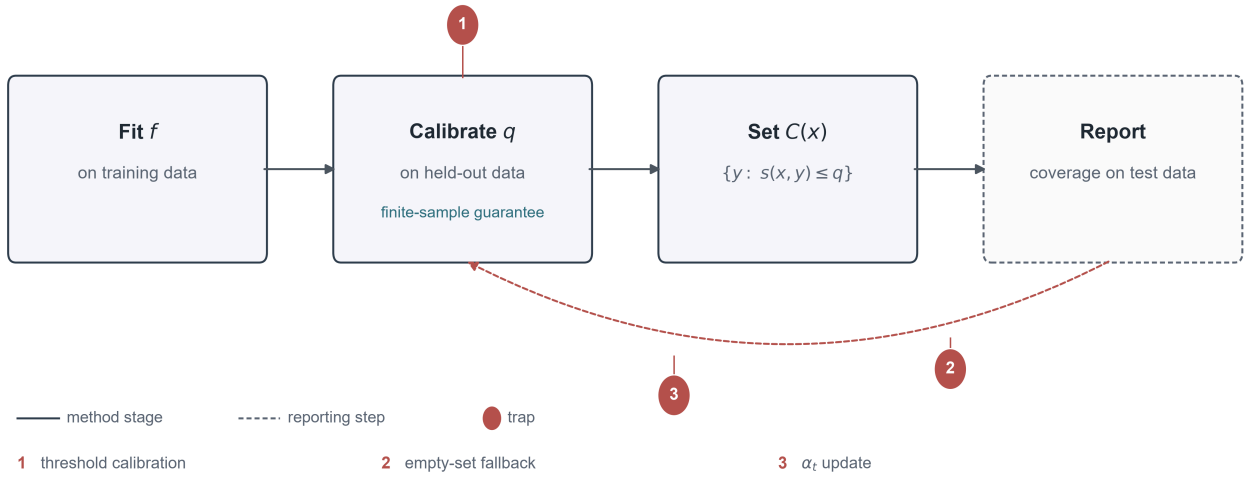
dataset	score	cells	mean empty rate	mean a_E	threshold c_a/c_e
digits	aps_rand	45	0.060	0.99	0.01
digits	aps_det	45	0.093	0.99	0.01
cifar10	aps_rand	60	0.020	0.96	0.04
cifar10	aps_det	45	0.084	0.99	0.01

dataset	score	cells	mean empty rate	mean a_E	threshold c_a/c_e
covertype	aps_rand	10	0.003	0.94	0.06
covertype	aps_det	10	0.003	0.94	0.06

The threshold is small in every dataset (0.01 to 0.06); the empty rate is not. The expected loss difference at a given cost ratio is proportional to the empty rate, so the same rule matters much more on digits than on covertype.

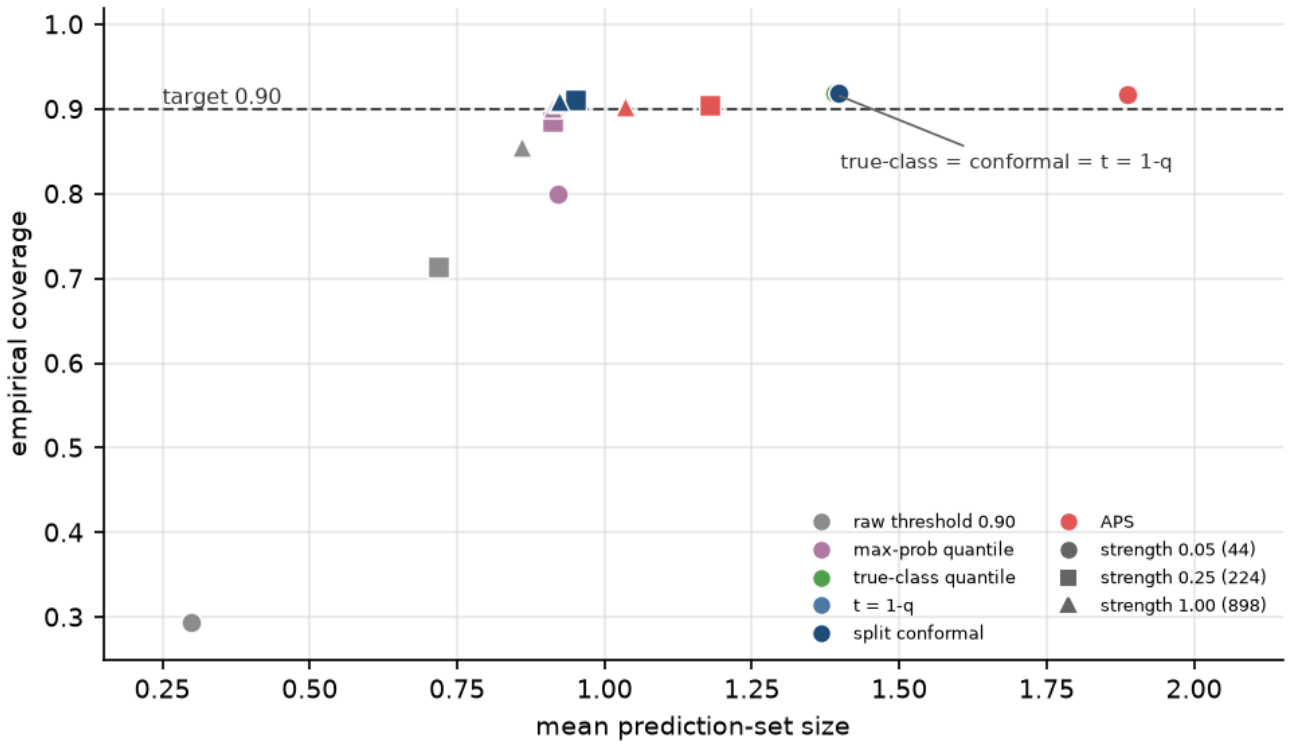
Figures

Figure 1



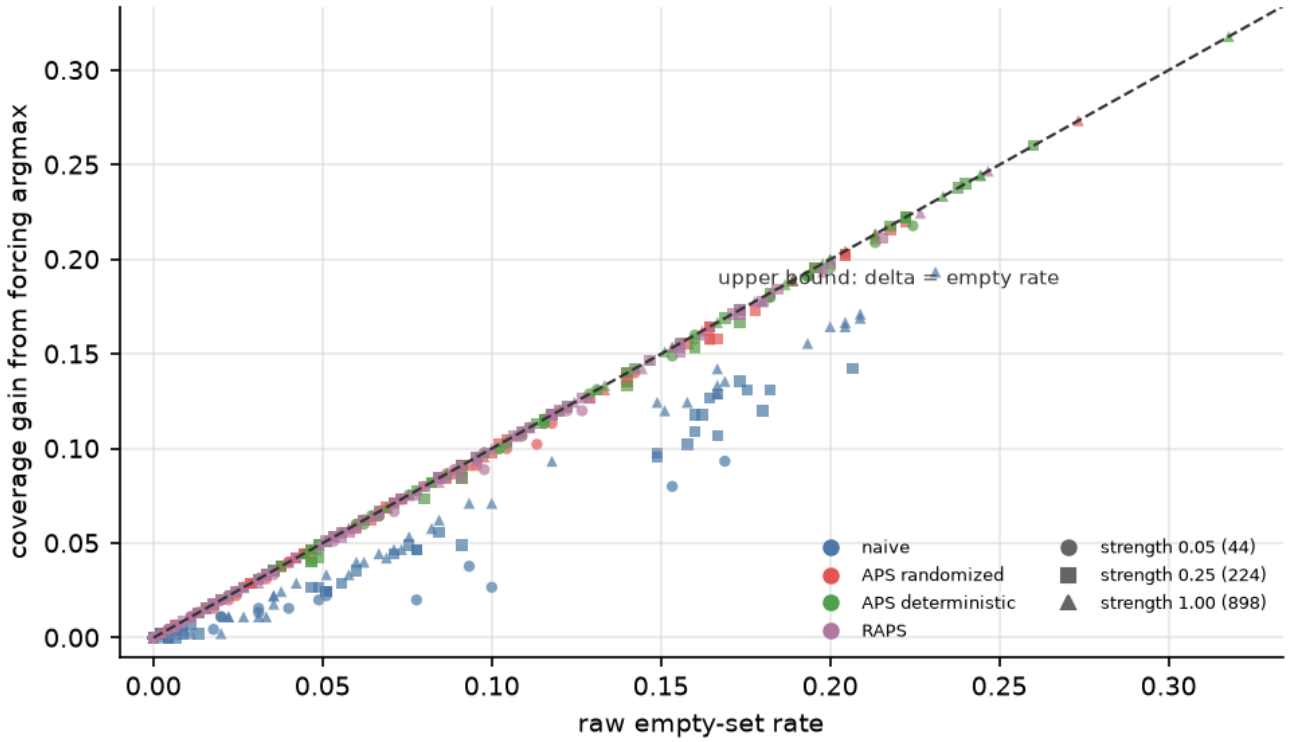
The three stages (solid) and the reporting step (dashed), with the three places where the reported number can move.

Figure 2



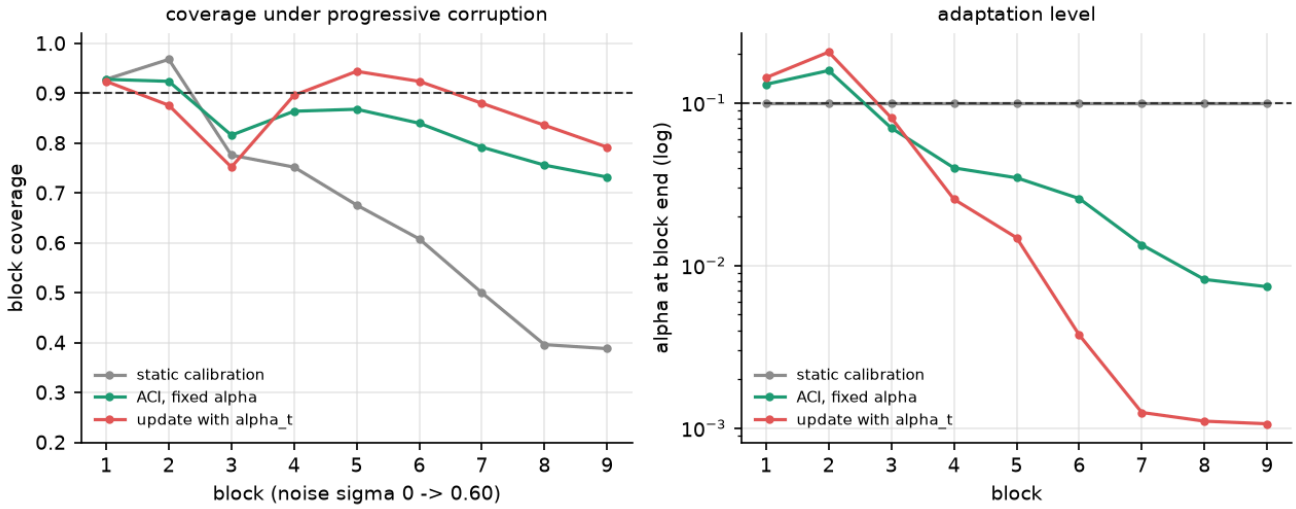
Coverage and set size versus the threshold on digits.

Figure 3



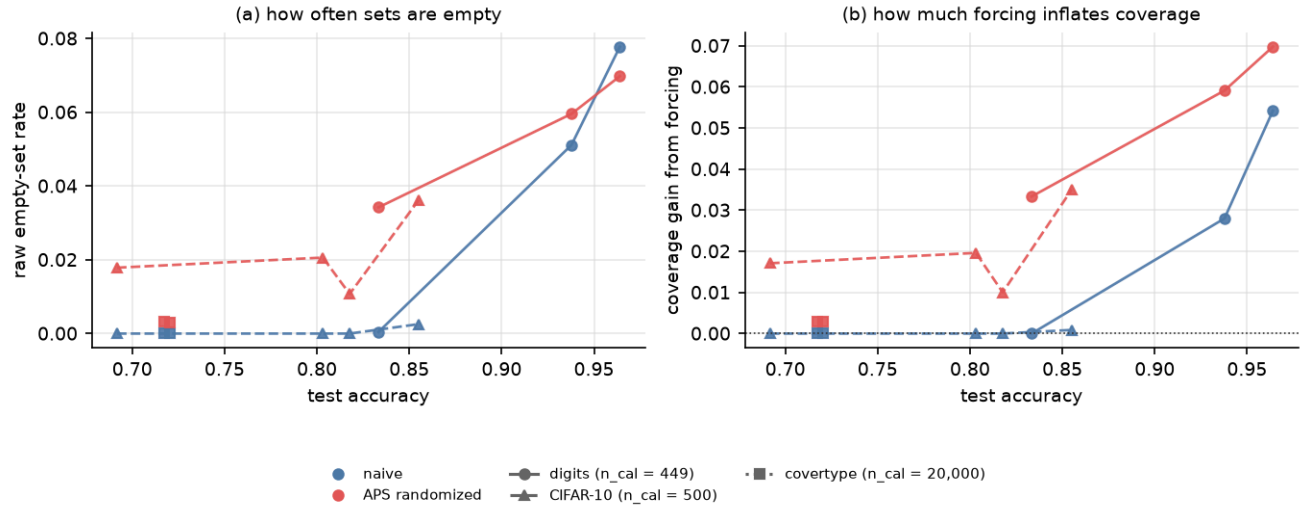
Change in coverage versus the raw empty rate across the image cells.

Figure 4



Coverage and level trajectories under drift on digits.

Figure 5



Empty rate and forcing gain versus test accuracy for digits, CIFAR-10 and covertype.

References

1. Aguru MK. What Split Conformal Prediction Guarantees, and What It Does Not: Exact Coverage Laws for Binary Classifiers, the Calibration-Set Size Needed for a Reliable Guarantee, and Class-Conditional Failure on UCI Adult. Zenodo (CERN European Organization for Nuclear Research), 2026. DOI 10.5281/zenodo.23582987. Preprint.
2. Al Bataineh A. Certified Label-Free Drift Alarms From Empty Conformal Prediction Sets for Metal-Oxide Gas Sensor Arrays. IEEE Sensors Letters, 2026. DOI 10.1109/lsens.2026.3748815.
3. Angelopoulos A N, Bates S. Conformal Prediction: A Gentle Introduction. Foundations and Trends in Machine Learning 16(4), 494-591, 2023. DOI 10.1561/2200000101.
4. Angelopoulos A N, Bates S, Candès E J, Jordan M I, Lei L. Learn then Test: Calibrating Predictive Algorithms to Achieve Risk Control. Annals of Applied Statistics 19(2), 1641–1662, 2025. DOI 10.1214/24-AOAS1998.
5. Angelopoulos A N, Bates S, Fisch A, Lei L, Schuster T. Conformal Risk Control. arXiv preprint, 2022. arXiv:2208.02814. Preprint.
6. Angelopoulos A N, Bates S, Malik J, Jordan M I. Uncertainty Sets for Image Classifiers using Conformal Prediction. International Conference on Learning Representations (ICLR 2021), 2021. arXiv:2009.14193.
7. Barber R F, Candès E J, Ramdas A, Tibshirani R J. The Limits of Distribution-Free Conditional Predictive Inference. Information and Inference 10(2), 455–482, 2021. DOI 10.1093/imaiai/iaaa017.
8. Barber R F, Candès E J, Ramdas A, Tibshirani R J. Conformal Prediction Beyond Exchangeability. Annals of Statistics 51(2), 816–845, 2023. DOI 10.1214/23-AOS2276.
9. Bates S, Angelopoulos A N, Lei L, Malik J, Jordan M I. Distribution-Free, Risk-Controlling Prediction Sets. Journal of the ACM 68(6), 1–34, 2021. DOI 10.1145/3478535.
10. Blackard J A. Covertype. UCI Machine Learning Repository, 1998. DOI 10.24432/C50K5N.
11. Boman SE. Benchmarking non-conformity score functions in conformal prediction. arXiv, 2026. arXiv:2605.24983. Preprint.
12. Campos M M, Farinhas A, Zerva C, Figueiredo M A T, Martins A F T. Conformal Prediction for Natural Language Processing: A Survey. Transactions of the Association for Computational Linguistics 12, 1497-1516, 2024. DOI 10.1162/tacl_a_00715.
13. Cohen K M, Park S, Simeone O, Shamai S S. Cross-Validation Conformal Risk Control. IEEE International Symposium on Information Theory (ISIT 2024), 250–255, 2024. DOI 10.1109/isit57864.2024.10619172.
14. Combes J, Coeurjolly J, Derville A. Macro Mondrian Conformal Risk Control. arXiv preprint, 2026. DOI 10.48550/arxiv.2610.35035. Preprint.
15. coverforest contributors. coverforest: conformal prediction with random forests. version 0.0.3, 2026. https://pypi.org/project/coverforest/.
16. crepes contributors. crepes: conformal regressors and predictive systems. version 0.9.1, 2026. https://pypi.org/project/crepes/.
17. Farinhas A, Zerva C, Ulmer D, Martins A F T. Non-Exchangeable Conformal Risk Control. arXiv preprint, 2023. arXiv:2310.01262. Preprint.
18. Fontana M, Zeni G, Vantini S. Conformal prediction: A unified review of theory and new challenges. Bernoulli 29(1), 1-23, 2023. DOI 10.3150/21-BEJ1447.
19. Gauthier E, Bach F, Jordan MI. Adaptive Coverage Policies in Conformal Prediction. arXiv preprint, 2025. DOI 10.48550/arxiv.2510.04318. Preprint.
20. Gibbs I, Candès E J. Adaptive Conformal Inference Under Distribution Shift. Advances in Neural Information Processing Systems 34 (NeurIPS 2021), 2021. arXiv:2106.00170.
21. Gibbs I, Cherian J J, Candès E J. Conformal Prediction with Conditional Guarantees. Journal of the Royal Statistical Society: Series B 87(4), 1100–1126, 2025. DOI 10.1093/jrsssb/qkaf008.

22. Hore R, Barber R F. Conformal Prediction with Local Weights: Randomization Enables Robust Guarantees. Journal of the Royal Statistical Society: Series B 87(2), 549–578, 2025. DOI 10.1093/jrsssb/qkae103.
23. Koul A. conformal: a minimal conformal prediction implementation for classification. version 0.1, 2017. https://pypi.org/project/conformal/.
24. Lei J, G'Sell M, Rinaldo A, Tibshirani R J, Wasserman L. Distribution-Free Predictive Inference for Regression. Journal of the American Statistical Association 113(523), 1094–1111, 2018. DOI 10.1080/01621459.2017.1307116.
25. Linusson H. nonconformist: Python implementation of the conformal prediction framework. version 2.1.0, 2017. https://github.com/donlnz/nonconformist.
26. MAPIE contributors. MAPIE: a scikit-learn-contrib package for conformal prediction. version 1.6.0, 2026. https://pypi.org/project/mapie/.
27. Mehrtens H, Bucher T, Brinker TJ. Pitfalls of Conformal Predictions for Medical Image Classification. arXiv, 2025. arXiv:2506.18162. Preprint.
28. Oliveira RI, Orenstein P, Ramos TR, Romano JV. Split Conformal Prediction and Non-Exchangeable Data. arXiv preprint, 2022. DOI 10.48550/arxiv.2203.15885. Preprint.
29. puncc contributors. puncc: predictive uncertainty calibration and conformal prediction. version 0.9.3, 2026. https://pypi.org/project/puncc/.
30. Romano Y, Patterson E, Candes E J. Conformalized Quantile Regression. Advances in Neural Information Processing Systems 32 (NeurIPS 2019), 2019. arXiv:1905.03222.
31. Romano Y, Sesia M, Candès E J. Classification with Valid and Adaptive Coverage. Advances in Neural Information Processing Systems 33 (NeurIPS 2020), 2020. arXiv:2006.02544.
32. Sadinle M, Lei J, Wasserman L. Least Ambiguous Set-Valued Classifiers With Bounded Error Levels. Journal of the American Statistical Association 114(525), 223–234, 2019. DOI 10.1080/01621459.2017.1395341.
33. Shafer G, Vovk V. A Tutorial on Conformal Prediction. Journal of Machine Learning Research 9, 371–421, 2008. DOI 10.5555/1390681.1390693.
34. Susmann H, Chambaz A, Josse J. AdaptiveConformal: an R package for adaptive conformal inference. arXiv preprint, 2023. arXiv:2312.00448. Preprint.
35. Tibshirani R J, Barber R F, Candès E J, Ramdas A. Conformal Prediction Under Covariate Shift. Advances in Neural Information Processing Systems 32 (NeurIPS 2019), 2019. arXiv:1904.06019.
36. TorchCP contributors. TorchCP: a Python library for conformal prediction. version 1.2.1, 2026. https://pypi.org/project/torchcp/.
37. Vovk V. Conditional validity of inductive conformal predictors. ACML 2012, PMLR 25, 475-490, 2012. https://proceedings.mlr.press/v25/vovk12.html.
38. Vovk V, Gammerman A, Shafer G. Algorithmic Learning in a Random World. Springer, New York, 2005. DOI 10.1007/b106715.
39. Zhou X, Chen B, Gui Y, Cheng L, Deng Z, Wang X, et al. Conformal Prediction: A Data Perspective. ACM Computing Surveys 58(2), 1–37, 2025. DOI 10.1145/3736575.